

A Passivity Obstruction: Why a Derived $a_0 = cH_\Lambda/Z$ and Single-Metric Weak-Lensing Cannot Coexist in de Sitter–Unruh Modified Inertia

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Abstract

The de Sitter–Unruh modified-inertia (MI) program reproduces galaxy dynamics through a passive vacuum-response kernel $K(\Box_u/a_0^2)$ whose bounded, causal, positive-measure (Herglotz–Nevanlinna) structure ties the MOND acceleration scale to the cosmological horizon, $a_0 = cH_\Lambda/Z$ with $Z = \sqrt{32\pi/3} = 5.789$. That structural tie is the program’s single most distinctive claim. **What the passive kernel fixes is the *scale* of a_0 ; its numerical value and the sign $s = -1$ of the coupling remain postulates — a one-parameter effective-field-theory reframing, not a first-principles value.** The interpolating ν -kernel law itself is Milgrom’s (1999); the framework’s distinctive content is the cH_Λ/Z coefficient (his was $2cH_\Lambda$) and the covariant MI completion built on the passive frame. Here we ask whether the *same* field theory can also reproduce the weak-lensing MOND relation — the phantom mass $(\nu - 1)\rho$ whose gravitational signature makes the observed lensing radial-acceleration relation coincide with the dynamical one, an equality that excludes single-metric pure MI at the $\sim 27\sigma$ level (Brouwer et al. 2021, conservative rail, both footings). Conditional on the kernel-passivity premise, it cannot, and we identify the obstruction exactly. Writing D (a_0 enters *solely* as the horizon-floored argument of the passive kernel), S (single-metric, $c_\gamma = c_{\text{GW}}$), G (ghost-free), L (the theory reproduces $g_{\text{lens}} = \nu g_{\text{bar}} \rightarrow \sqrt{a_0 g_{\text{bar}}}$, null geodesics feeling the phantom), the central implication is $D \wedge S \Rightarrow \neg L$. Equivalently $\{D, S, L\}$ is the *unique minimal unsatisfiable subset* of $\{D, S, G, L\}$: each proper subset is realized — $\{D, S\}$ is pure MI, $\{S, L\}$ is the Deffayet–Esposito–Farèse–Woodard / AeST class with a_0 a free coupling, and $\{D, L\}$ is a disformal two-cone construction that keeps a_0 derived but is excluded observationally by GW170817 (violating S). Ghost-freedom G enters no step. The mechanism is a *passivity/amplification dichotomy*: the kernel derives a_0 ’s scale only because it is passive ($\sup K = 1$), and a passive kernel can only *suppress* the source to ρ/ν , whereas MOND-lensing demands *enhancement* to $\nu\rho$, strictly outside the passive cone. We are explicit that the obstruction is conditional on that passivity premise — but the premise is precisely what derives a_0 , so within the passive class the derivation and $\neg L$ are consequences of one hypothesis; whether any *non-passive* (pumped) kernel could derive a_0 is a separate open question that the banked sign walls shut pump-free. All a_0 -valued numbers are reported on both footings ($9.36 \times 10^{-11} \text{ m s}^{-2}$ canonical and 1.13×10^{-10} alternate); the verdict is footing-independent. This is a conditional obstruction, not a closure of the program.

1. Introduction

1.1 The framework

Modified inertia (MI) treats MOND phenomenology not as a modification of the gravitational field sourced by matter, but as a modification of the inertial response of matter to a field. In the de Sitter–Unruh realization, a passive cosmic rest frame u^μ — a unit-timelike vector carrying **zero** propagating degrees of freedom, fixed by a Lagrange constraint $(u \cdot u + 1) = 0$ — mediates a vacuum response whose kernel acts on the frame d’Alembertian $\square_u$. The worldline equation of motion that emerges is

$$g_{\text{obs}} = \nu(y) g_{\text{bar}}, \quad \nu(y) = \sqrt{1 + \frac{1}{y}}, \quad y \equiv \frac{g_{\text{bar}}}{a_0},$$

reducing to the deep-MOND law $g_{\text{obs}} \rightarrow \sqrt{a_0 g_{\text{bar}}}$ as $y \rightarrow 0$ and to Newton as $y \rightarrow \infty$. This fits the SPARC rotation-curve sample at 0.108 dex scatter ($\Upsilon = 0.70$), competitive with regularized MOND.

The distinguishing structural claim is the identity of the acceleration scale with a horizon quantity,

$$a_0 = \frac{cH_\Lambda}{Z}, \quad Z = \sqrt{\frac{32\pi}{3}} = 5.789.$$

Throughout, *footing* denotes the choice of cosmological input to this expression: the **canonical** pure- Λ footing ($\rho_{\text{DE}}/cH_\Lambda$) gives $a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$, and the **alternate** footing (ρ_{total}/cH_0) gives 1.13×10^{-10} . Both are carried at every a_0 -valued number below.

1.2 The kernel and its passivity premise

The tie of a_0 ’s scale to the horizon arises because the vacuum-response kernel is *passive*: it is a bounded, causal, positive-measure response admitting a Herglotz–Nevanlinna spectral representation, normalized so that $\|K\| \leq 1$ with a sum rule $\int d\mu/|t| = 1$ (banked MI action, DOI 10.5281/zenodo.21264727; v11 21284144). Passivity — a positive spectral measure with unit total weight — is what floors the response amplitude at the horizon scale cH_Λ ; it is the source of the scale-tie. We take this passivity as the **single premise** of the present result and are explicit (Section 7) that the obstruction is conditional on it.

The real-space object that enters the static weak-field problem is the on-shell dressing. In the quasistatic first-moment reduction $K(\square_u/a_0^2) \rightarrow K(X)$ with $X = |a|^2/a_0^2$ and $a^\mu = u^b \nabla_b u^\mu$, the interpolating kernel is

$$K(X) = \frac{\sqrt{1+4X}-1}{2\sqrt{X}}, \quad \nu = 1/K.$$

This $K(X)$ is a **bounded, increasing (Bernstein-type) function**: $K(0^+) = 0$ at the deep-MOND end and $\sup_{X \geq 0} K(X) = 1$ saturated as $X \rightarrow \infty$ (Newton), with $\nu = 1/K$ running from ∞ (deep-MOND) down to 1 (Newton). (It is *not* completely monotone — completely-monotone functions decrease; here K increases and K' is completely monotone, the Bernstein/Stieltjes dual. The banked Herglotz–Nevanlinna, $\|K\| \leq 1$ and sum-rule properties attach to the *frequency-domain response kernel* $K(\square_u)$; the static $K(X)$ is the corresponding real-space dressing.) The single fact the obstruction actually uses is the uniform bound

$$\boxed{\sup_{X \geq 0} K(X) = 1} \quad (\text{Newtonian saturation; } K \rightarrow 0 \text{ deep-MOND}),$$

i.e. the passive kernel can only ever *suppress* a source to $\rho K \leq \rho$.

1.3 Honest scope of the claim

We are precise about what is and is not derived. What the framework establishes is the **structural tie of the scale of a_0** to the passive kernel and the de Sitter horizon. The **numerical value** of a_0 and the **sign** $s = -1$ of the coupling remain **postulates** — the program is a one-parameter effective-field-theory *reframing*, not a first-principles computation. Moreover, the ν -kernel law itself is not original: it is Milgrom (1999, *Phys. Lett. A* **253**, 273; astro-ph/9805346, Eq. 9), whose modified-inertia formulation uses an identical interpolating kernel (his coefficient was $2cH_\Lambda$). The framework’s distinctive content is precisely (i) the cH_Λ/Z coefficient and (ii) the covariant MI completion built on the passive frame. We credit Milgrom’s original MOND (1983) and his 1999 modified-inertia formulation explicitly, and we do not re-assert any earlier, retracted claims of a derived value of a_0 , a Standard-Model bridge, or a final all-physics theory. The defensible position is the a_0 reframing, and nothing more.

1.4 The lensing question

Galaxy dynamics is a worldline phenomenon; weak lensing is not. Lensing probes the deflection of null geodesics, fixed by the gravitating stress tensor $T_{\mu\nu}$ that the field equations actually see. A prior calculation (the single-metric lensing decider; SOLVE.md in the banked `mi_lensing_final/` folder, folded into MI Field Theory Results 2026, DOI 10.5281/zenodo.21403470, erratum 21415677) assembled the **total** gravitating stress tensor of the MI action — matter legs and frame-constraint legs together — and found that it sources a *suppressed* effective density $\rho_{\text{eff}} = \rho K = \rho/\nu$. The single metric therefore under-lenses: the lensing ratio

$$F(y) \equiv \frac{g_{\text{lens}}}{\nu(y) g_{\text{bar}}} < \frac{1}{\nu(y)} < 1 \quad \text{everywhere.}$$

The observed relation is different. The measured weak-lensing radial-acceleration relation (RAR) of Brouwer et al. (2021, KiDS-1000 isolated galaxies) coincides with the *dynamical* RAR: null geodesics feel an effective lensing mass enhanced to $\nu\rho$, not suppressed to ρ/ν . Single-metric pure MI is excluded relative to that lensing-RAR = dynamical-RAR equality at $\sim 27\sigma$ (conservative rail, footing-independent). The gap between what is sourced (ρ/ν) and what the equality requires ($\nu\rho$) is a factor ν^2 .

1.5 Statement of the result

The natural follow-up is a completion question: can the ν^2 wedge be closed by adding structure to the action, while retaining everything that makes the program distinctive? We formalize “everything distinctive” as four desiderata (Section 3) and show that, **conditional on kernel passivity**, the target is unsatisfiable. The result is an **obstruction theorem**:

Assume kernel passivity. Then $D \wedge S \Rightarrow \neg L$: no field theory extending the MI action keeps a_0 tied to the passive kernel (D), propagates photons and gravitons on one metric (S), and reproduces the single-metric MOND-lensing phantom $(\nu - 1)\rho$ (L). Equivalently $\{D, S, L\}$ is the unique minimal unsatisfiable subset of $\{D, S, G, L\}$; ghost-freedom G enters no step.

The physical crux is a passivity/amplification dichotomy (Section 6): the property that derives a_0 ’s scale (passivity, $\sup K = 1$) is the very property that forbids the lensing enhancement (which needs $\nu > 1$, outside the passive cone). A completion *does* exist — but only as modified *gravity*, at the cost of a_0 becoming a free coupling. We frame this as neither a triumph nor a funeral: it is an exact obstruction, conditional on passivity, that shows the a_0 reframing is load-bearing, and it redirects the search for an MI-versus-MG discriminant away from lensing (now a shared observable) and back to the relational velocity-dispersion spread.

2. Setup

2.1 The action and its kernel

The covariant MI action is (signature $-+++$)

$$S = S_{\text{EH}}[g] + S_u[g, u, \lambda] + S_{\text{matter}}[g, u, \psi; K],$$

with S_{EH} the Einstein–Hilbert term, $S_u = -\int \sqrt{-g} \frac{\lambda}{2} (u \cdot u + 1)$ the passive frame with its Lagrange multiplier (zero propagating dof), and a matter sector dressed by the kernel,

$$S_{\text{matter}} \supset -\frac{1}{2} \int \sqrt{-g} \rho_m s (u \cdot u) K(\Box_u/a_0^2), \quad s = -1.$$

The passivity premise (Section 1.2) attaches to $K(\Box_u)$: bounded, causal, positive spectral measure, $\|K\| \leq 1$, sum rule $\int d\mu/|t| = 1$ (banked v4, v11). These properties underwrite two things at once — the causal-retarded well-posedness of the theory, and the identification $a_0 = cH_\Lambda/Z$ (the amplitude of a passive vacuum response is floored at the horizon scale). The static real-space dressing $K(X)$ of Section 1.2 is the object that enters the weak-field computation; the load-bearing fact we use downstream is only $\sup K = 1$.

2.2 The worldline law

On the rotation-curve shell the kernel argument collapses. With $X = |a|^2/a_0^2$ and the on-shell condition $|a| = g_{\text{obs}} = \nu g_{\text{bar}}$ (so $|a|/a_0 = \nu y = \sqrt{y^2 + y}$, hence $X = y^2 + y$), one has $1 + 4X = 1 + 4(y^2 + y) = (2y + 1)^2$, and the radical collapses:

$$K|_{\text{on-shell}} = \frac{(2y + 1) - 1}{2\sqrt{y^2 + y}} = \frac{1}{\nu(y)} \text{ exactly,} \quad \left. \frac{2K'X}{K} \right|_{\text{on-shell}} = \frac{1}{2y + 1} \leq 1.$$

The on-shell dressing is the *suppression* $1/\nu \leq 1$, and the anisotropic slip factor $2K'X/K = 1/(2y + 1)$ is bounded and tension-signed.

2.3 The assembled stress tensor and the ν^2 wedge

Assembling matter and frame legs in the only Newton-consistent bookkeeping (the composite frame $u = J^\mu/|J|$ for which $u \cdot u = -1$ identically, so the frame-constraint leg $S_u \equiv 0$ and its stress $T^u \equiv 0$; the $1/\nu$ dressing therefore comes entirely from the **matter** leg ρK , not from a frame leg) yields

$$\hat{T}_{\mu\nu} = \rho K(X) u_\mu u_\nu - 2 \frac{\rho K'(X)}{a_0^2} a_\mu a_\nu.$$

On the RAR shell this sources $\rho_{\text{eff}} = \rho K = \rho/\nu$ (a *suppression*) plus an anisotropic radial tension $\Pi = -\rho_{\text{eff}}/(2y + 1)$ carrying the bounded, tension-signed slip factor — it *reduces* lensing rather than enhancing it. Linearizing Einstein on the single metric $(ds^2 = -(1 + 2\Phi)dt^2 + (1 - 2\Psi)d\vec{x}^2)$,

$$\nabla^2 \Psi = 4\pi G \rho_{\text{eff}}, \quad \Phi' - \Psi' = 4\pi G r \Pi, \quad g_{\text{lens}} = \frac{1}{2}(\Phi' + \Psi'),$$

every source term is $O(K) \leq 1$. There is **no** $O(\nu)$ **structure anywhere** in the assembled tensor.

The **wedge** is the gap between what lensing needs and what MI sources. The observed lensing-RAR = dynamical-RAR equality needs effective mass $\nu\rho$ (enhanced *up*), while MI supplies ρ/ν (suppressed *down*):

$$\underbrace{\nu\rho}_{\text{needed}} \quad \text{vs} \quad \underbrace{\rho/\nu}_{\text{sourced}} \quad \Rightarrow \quad \text{factor } \nu^2 = 1 + \frac{1}{y}.$$

The missing piece is the **phantom mass** $M_{\text{ph}} = (\nu - 1)M_{\text{bar}}$, exactly the density that QUMOND/AeST source gravitationally and MI does not. The whole completion problem is: *can the phantom be sourced without giving up the derivation of a_0 's scale?*

3. The completion problem

3.1 The desiderata

A satisfying MI completion of lensing would keep all of the following. We state each narrowly, so the theorem rests on definitions rather than on plausibility steps.

- **D (a_0 -derived).** a_0 enters the theory *solely* as the horizon-floored argument of the passive kernel $K(\Box_u/a_0^2)$ — **not** as an independent Lagrangian coupling. (D is the *scale-tie*; the value and sign remain postulates, Section 1.3.)
- **S (single-metric).** Photons and gravitons propagate on one metric, $c_\gamma = c_{\text{GW}}$.
- **G (ghost-free).** No Ostrogradsky mode, no negative-norm state.
- **L (MOND-lensing).** The theory reproduces $g_{\text{lens}} = \nu(y) g_{\text{bar}} \rightarrow \sqrt{a_0 g_{\text{bar}}}$ ($F \rightarrow 1$): null geodesics feel the phantom $(\nu - 1)\rho$. *We state L without reference to the number of metrics* — this is what makes S a non-redundant, essential hypothesis (Section 5).

with two secondary constraints — **Cassini** (Solar-System PPN safety, $\nu - 1 \sim 1/(2y) \rightarrow 0$ at $y \sim 10^6$) and **cosmology-preservation** (the growing mode $\nu_{\text{cosmo}} \in [1, 1.09]$, $\sigma_8 + 2\text{--}3\%$, from the horizon-floored argument).

3.2 Two lemmas

The theorem is carried by two universal lemmas; the candidate constructions of Section 4 are existence-witnesses for the surviving corners, not the argument itself.

Lemma 1 (nonlocality — uses S). Under S , the lensing deflection is set by the single metric's stress tensor, so L requires the phantom $(\nu - 1)\rho$ to

be **real gravitating stress-energy**. For a point mass M ($g_{\text{bar}} = GM/r^2$, $y = GM/(a_0 r^2)$) the enclosed phantom that makes $g_{\text{lens}} = \nu g_{\text{bar}}$ is $M_{\text{ph}}(r) = (\nu(y) - 1)M$, and (sympy-exact)

$$\frac{M_{\text{ph}}(r)}{r} \xrightarrow{r \rightarrow \infty} \sqrt{\frac{a_0 M}{G}} \neq 0,$$

an **unbounded halo** whose enclosed mass keeps growing where $\rho = 0$. Now the precise content: a *local* functional of ρ and the passive frame produces a phantom whose radial profile is fixed by the local acceleration invariant $|a|(r)$ **alone**, so its amplitude cannot carry the source mass M that $(\nu - 1)M$ requires — the same functional applied to two different masses gets the profile wrong for at least one (this is the mass-blindness that kills C1, §4.1). Matching the *mass-correct, universal* phantom for **all** sources is therefore what requires a nonlocal carrier — a term whose variation yields the elliptic equation $\nabla \cdot [\dots] = \rho$ (the inverse-Laplacian of AQUAL/QUMOND). (A local term can produce a nonzero off-source phantom; what it cannot produce is the mass-correct one for every body.)

Lemma 2 (boundedness — uses passivity). By $\sup K = 1$, the passive kernel delivers only $O(K) \leq 1$ dressing. On the RAR shell it gives the *uu*-coefficient $K = 1/\nu$ (suppression) and slip $2K'X/K = 1/(2y + 1)$ (bounded, tension). The phantom instead needs *uu*-coefficient $\nu - 1/\nu = 1/\sqrt{y(y + 1)}$ (deep-MOND $\sim 1/\sqrt{y}$, unbounded); the shortfall

$$\frac{\text{needed}}{\text{supplied}} = \frac{\nu - 1/\nu}{1/\nu} = \nu^2 - 1 = \frac{1}{y}$$

diverges as $y \rightarrow 0$. No normalized passive kernel reaches ν on the RAR shell (a direct counterexample search finds the passive cap 1.0 against a deep-MOND target ≈ 31.6 ; the value the frame actually realizes, $1/\nu$, is the *farthest* point from ν).

Both footings: the halo amplitude $\sqrt{a_0 M/G}$ carries a_0 , but the *nonlocality* (the nonzero limit) and the shortfall $1/y$ are footing-independent — they hold for 9.36×10^{-11} and 1.13×10^{-10} identically.

4. Candidate mechanisms

We computed three explicit completions. Each is a genuine attempt to close the wedge; each is an existence-witness landing on a different corner of the obstruction. All scripts exit 0 on both footings.

4.1 C1 — nonminimal frame-curvature: keeps D , fails L by mass-blindness

Add a frame-kernel-curvature coupling with **no new dimensionful constant**,

$$\Delta S = \frac{c^4}{16\pi G} \int \sqrt{-g} F(K(\Box_u/a_0^2)) Q, \quad Q \in \{R, u^\mu u^\nu R_{\mu\nu}\}.$$

Because a_0 enters only through X inside K and the prefactor is the bare $c^4/16\pi G$, this is a **clean D -side witness**: the coupling is dimensionless in F , and a_0 appears nowhere except as the horizon-floored kernel argument. (A separate dissatisfaction — that the *shape* of the free function F is not itself fixed by the vacuum — is not an a_0 -scale freedom and, as we now show, is not what defeats the candidate; L fails first, on mass-blindness.)

Scalar-Ricci variation gives the standard scalar-tensor form; the new gravitating source is $-(g_{\mu\nu}\Box - \nabla_\mu \nabla_\nu)F(K)$, whose static 00-projection is a Laplacian of a local scalar. By Gauss the enclosed phantom is a pure surface flux, $M_{\text{ph}}(r) = (c^2/G) \kappa r^2 F'(r)$ — nonzero off-source, so a local term *does* source a phantom. But imposing lensing closure $M_{\text{ph}} = (\nu - 1)M_{\text{bar}}$ on a point mass yields (sympy-exact)

$$F'(y) = -\frac{\nu(y) - 1}{2\kappa\sqrt{y}} \frac{\sqrt{GMa_0}}{c^2} \propto \sqrt{M}.$$

The required shape **carries the source mass**: $F'_{\text{req}}(y; M_2)/F'_{\text{req}}(y; M_1) = \sqrt{M_2/M_1}$, independent of y . A local $F(K)$ sees only $|a|$ — it is *blind to the source mass* — so no single universal F sources the phantom for more than one object (Lemma 1). Fixing F on a $6 \times 10^{10} M_\odot$ galaxy and applying it to a $10^{14} M_\odot$ cluster, the cluster receives phantom fraction

$$\sqrt{\frac{6 \times 10^{10}}{10^{14}}} \approx 0.024,$$

i.e. $\sim 2.4\%$ of the needed phantom — under-lensed by a factor $1/0.0245 \approx 41$. By the $\sqrt{}$ symmetry the reverse tuning (fix on the cluster, apply to the galaxy) over-lenses by the same $\approx 41\times$. The miss is footing-independent (a_0 cancels in the ratio). Separately, under honest metric variation $X = |a|^2/a_0^2$ is a function of first metric derivatives, so $\Box F(K)$ contains third metric derivatives (Φ'') — a genuine Ostrogradsky ghost, unless one *freezes* X and prescribes the scalar by hand (which is no longer MI). Crucially, **C1 fails L by mass-blindness regardless of its ghost**, so it is a clean $D = 1$ witness for the “ G is not the lever” step. Verdict: **fails-lensing, D kept**.

4.2 C2 — nonlocal-in-matter (Deffayet–Esposito–Farèse–Woodard class): closes L , loses D

Take a purely-metric, single-metric, nonlocal-in-matter distortion of Einstein–Hilbert,

$$S = \frac{1}{16\pi G} \int \sqrt{-g} R[1 + f(\Box^{-1}R)] + S_{\text{matter}}[g, \psi],$$

with $\Box^{-1}$ a *retarded* Green function (so the distortion is a functional of matter history) and f chosen so the weak-field limit is the QUMOND/AQUAL modified-Poisson equation $\nabla \cdot [\nu(|\nabla\Phi_N|/a_0)\nabla\Phi_N] = 4\pi G\rho$. This is the nonlocal-metric MOND realization of Deffayet, Esposito-Farèse & Woodard (2011).

It works. The nonlocal term sources the phantom mass $\nu\rho$ on the single metric as a genuine scalar energy density, with **zero** anisotropic stress; the no-slip theorem then gives $\Phi = \Psi$, so $g_{\text{lens}} = \nu g_{\text{bar}}$ and the deep slope $\rightarrow \sqrt{a_0 g_{\text{bar}}}$ ($F \rightarrow 1$, machine-checked). It is purely metric, so $c_\gamma = c_{\text{GW}}$ (GW170817-safe), and Cassini-safe by construction ($\nu - 1 = 5 \times 10^{-7}$ at $y = 10^6$).

But a_0 is now **free**, and the load-bearing reason is *structural*, not merely a fit degeneracy: there is **no passive frame**, **no kernel** $K(\Box_u)$, **no** $Z = \sqrt{32\pi/3}$ — the nonlocal operator acts on R (matter curvature), not on a vacuum frame, so nothing ties a_0 to cH_Λ . It enters as an independent coupling inside f , i.e. $\neg D$ by definition. (Corroborating, but weaker: the MOND relation $g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + a_0 g_{\text{bar}}}$ is form-invariant under $(a_0, g_{\text{bar}}) \rightarrow \lambda(a_0, g_{\text{bar}})$, so both footings fit equally. This footing-non-diagnosticsity is a property of the RAR *relation itself* and is **shared by a_0 -derived pure MI** — the SPARC RAR is likewise convention-compatible and non-diagnostic of 9.36×10^{-11} — so it does not by itself mark C2 as MG; the structural absence of the kernel does.) The one residual partial: $\Box^{-1}$ carries a horizon IR scale, so $a_0 \sim cH_\Lambda$ is *natural* parametrically — but natural is not derived, and Z (hence the exact value) is not sourced. Secondary costs: a localized would-be ghost (kinetic eigenvalues $\pm \frac{1}{2}$) tamed only by the *contested* retarded prescription, and a separate cosmology fit of f . Verdict: **closes L , a_0 free (= MG)**.

4.3 C3 — frame/scalar carrier: closes L , loses D , plus one new dof

Ask whether the frame’s dressed momentum can source curvature. Sub-candidate **C3a** (passive frame only, a_0 kept derived, 0 new dof): the matter leg’s uu coefficient is $O(K) = 1/\nu$ and the frame-constraint leg vanishes identically ($u \cdot u = -1$), so the assembled source delivers the suppression $1/\nu$, short of the needed ν by $\nu^2 - 1 = 1/y$, diverging deep-MOND. **Fails lensing**, reproducing the SOLVE.md result exactly. Sub-candidate **C3b** (promote the carrier to a nonlocal scalar): the only local $h(X)$ reaching the deep phantom is $h \sim a_0/|a|$ — the AQUAL term — which makes the potential equation **nonlinear** and the phantom nonlocal (the same unbounded halo $M_{\text{ph}}/r \rightarrow \sqrt{a_0 M/G}$).

A spatial-kinetic carrier is then mandatory, so it **propagates**: one new scalar dof, its acceleration scale a free coupling. Lensing closes ($F \rightarrow 1$), $c_\gamma = c_{\text{GW}}$ holds (the scalar sources via its own $T_{\mu\nu}$, no disformal photon coupling), ghost-free holds ($f' > 0$), Cassini-safe — but a_0 is **free** and cosmology is forfeited to the new scalar. Verdict: **closes** L , a_0 **free** (= **MG**), +1 **scalar dof**.

4.4 Scorecard

lane	L candidate ΔS	S (lens- ing) $(c_\gamma = c_{\text{GW}})$	G (ghost) G	Cassini	cosmology	a_0	verdict
C1	$\int \sqrt{-g} F(R, u/R, uuR_{\mu\nu})$ (mass-blind $\sqrt{M}$; 2.4% / $41\times$)	FAIL (mass-blind $\sqrt{M}$; 2.4% / $41\times$)	FAIL (scalar- R) / risk $(uuR_{\mu\nu})$ grad-sky $\Phi''')$	OK*	plausible	kept- derived	fails- lensing
C2	$\int \sqrt{-g} R[f(\Box^{-1}R)]$ (re-tarded slope)	PASS (purely met-ric)	PASS (cost $(\pm \frac{1}{2};$ prescription-dependent)	PASS*	tune (sepa-rate fit)	FREE	closes, a_0 free
C3a	passive $-\frac{1}{2}\rho h(X)$ (deliv-ers $1/\nu$, short $1/y$)	FAIL (deliv-ers $1/\nu$, short $1/y$)	PASS (0 dof)	PASS	PASS	kept- derived	fails- lensing
C3b	freed nonlo-cal carrier (a_0/ a)	PASS ($F \rightarrow 1$, $\sqrt{\text{slope}}$)	PASS (own $T_{\mu\nu}$)	PASS ($f' > 0$)	forfeited	FREE (+1 scalar dof)	closes, a_0 free

The pattern is exact: every lane that keeps D (C1, C3a) fails L ; every lane that achieves L (C2, C3b) loses D . No lane holds both. Note C2's G -bit is *prescription-dependent* (whether the $\pm \frac{1}{2}$ block is a genuine ghost or merely non-propagating hinges on the contested retarded prescription); C3b is the clean ghost-free L -side witness. Either way C2 lands on $\neg D$ independent of its G -bit. These are witnesses; the rigor of the result lives in Lemmas 1–2 (Section 5), not in the enumeration.

5. The obstruction theorem

5.1 Statement

Theorem (conditional on kernel passivity). *Assume the passivity premise (Section 1.2). Then $D \wedge S \Rightarrow \neg L$: no field theory extending the MI action keeps a_0 as the passive-kernel argument (D), propagates on one metric (S), and reproduces the MOND-lensing phantom $(\nu - 1)\rho$ (L). Equivalently, $\{D, S, L\}$ is the **unique minimal unsatisfiable subset** of $\{D, S, G, L\}$: each proper subset is realized —*

- $\{D, S\}$ = pure MI (the current theory; L off);
- $\{S, L\}$ = Deffayet–Esposito–Farèse–Woodard / AeST with a_0 a **free** coupling (D off);
- $\{D, L\}$ = a disformal two-cone construction $\tilde{g} = g + B(\Box_u) u_\mu u_\nu$ with B in the passive kernel, which keeps a_0 derived but carries a second photon cone excluded observationally by GW170817 (S off).

Ghost-freedom G enters no step.

The connective is “**cannot coexist,**” $\neg(D \wedge L)$ **under S** — not exclusive-or: the pure-GR corner has *neither* D nor L , which an XOR would misclassify. Earlier notes that wrote “ $D \oplus L$ / XOR” are corrected here.

5.2 Why S is essential (and the theorem non-redundant)

Because L is stated metric-agnostically (Section 3.1), S is a genuine, independent hypothesis with a precise role: **under S , lensing is fixed by the one metric’s $T_{\mu\nu}$, so L forces the phantom to be real gravitating stress-energy**, and Lemma 1 then bites. Drop S — allow a disformal second cone $\tilde{g} = g + B u u$ — and photons see a *different* metric: the phantom becomes **pure geometry with no new stress-energy**, Lemma 1 is evaded, a_0 stays inside the passive kernel $B(\Box_u)$, and L is achieved with D kept. That $\{D, L, \neg S\}$ construction is logically satisfiable as a field theory; it is killed only *observationally*, by GW170817: a disformal coupling large enough to carry the $O(\nu)$ lensing deflection splits the photon and graviton cones far above the bound $|c_\gamma/c_{\text{GW}} - 1| \lesssim 10^{-15}$ (Abbott et al. 2017), while the GW-safe small- B regime cannot source an $O(\nu)$ deflection — the same passivity/amplification wedge, now on the disformal axis. (A merely *conformal* bimetric $\tilde{g} = A(\phi)g$ preserves the null cone, $c_\gamma = c_{\text{GW}}$, but cannot bend light beyond its baryonic content at all — the historical reason TeVeS (Bekenstein 2004) needed a disformal/vector part; conformal rescaling is a distinct non-escape.) This is exactly why S belongs in the minimal unsatisfiable subset: the $\{D, L\}$ witness exists, and S is what excludes it.

5.3 The mechanism as lemmas, and the enumeration as bookkeeping

The theorem’s content is the single implication $D \wedge S \Rightarrow \neg L$, carried by:

- **Lemma 1 (nonlocality, §3.2, uses S):** a mass-correct universal single-metric L requires a nonlocal $(\nu - 1)\rho$ carrier.
- **Corollary (propagation $\Rightarrow \neg D$):** a nonlocal carrier reproducing the elliptic phantom has a spatial-kinetic term, so it propagates (a new dof) or its shape is a free function put in by hand; either way its acceleration scale is an independent Lagrangian coupling, i.e. **not** the passive-kernel argument — $\neg D$ **by the definition of D** (this is where narrow- D makes the step a near-tautology rather than a plausibility claim). [witnesses: C2, C3b]
- **Lemma 2 (boundedness, §3.2, uses passivity):** keeping D confines the modification to the passive kernel $\sup K = 1$ ($O(K)$ dressing, $T_{00} = 0$ off $\text{supp } \rho$), which cannot source the $O(\nu)$ unbounded phantom — $\neg L$. [witnesses: C1, C3a]

A finite truth table over the 16 assignments of $\{D, S, G, L\}$ (script `nogo.py`, exit 0) then reads off that the target $D = S = L = 1$ is absent, that $\{D, S, L\}$ is the unique minimal unsatisfiable subset, and that the $\{D, L\}$ witness survives the mechanism but is GW-excluded. **We emphasize the table is bookkeeping:** it is modus ponens over Lemmas 1–2, and the entire physics content lives in those two lemmas over the examined term classes (frame-curvature, nonlocal-in-matter, frame carrier). It is not an exhaustion over all possible Lagrangians, and the result’s strength is exactly the strength of Lemmas 1–2 as physics statements over those classes.

The observationally-viable single-metric corners are:

D	S	G	L	identity
1	1	1	0	pure MI / C3a — a_0 derived, ghost-free, under-lenses ($F < 1/\nu$; the current theory)
1	1	0	0	C1 — local ghostly frame term; still fails L (mass-blind) even with a ghost

D	S	G	L	identity
0	1	1	1	C2, C3b — closes L , single-metric, ghost-free, but a_0 free (= MG)
0	1	0	1	nonlocal with untamed ghost — closes L , a_0 free, ghost (contested)
0	1	1	0	GR / trivial
0	1	0	0	GR + ghost (vacuous)

The target $D = S = G = L = 1$ is **absent**, and so is its ghost-relaxed sibling $D = S = 1, G = 0, L = 1$: relaxing ghost-freedom does not rescue $\{D, S, L\}$, because C1 fails L by mass-blindness regardless of its ghost. This is why G is not the lever.

5.4 Footing independence

The wedge is a_0 -free. The shortfall $\nu^2 - 1 = 1/y$ and the mass-blindness ratio $\sqrt{M_2/M_1}$ are dimensionless in $y = g_{\text{bar}}/a_0$, and ν is form-invariant under $(a_0, g_{\text{bar}}) \rightarrow \lambda(a_0, g_{\text{bar}})$. Both footings (9.36×10^{-11} / 1.13×10^{-10}) give the identical dimensionless verdict.

6. The passivity/amplification dichotomy

Here is the physical crux, the intended central idea of the paper.

$a_0 = cH_\Lambda/Z$ is derivable **only because** the MI modification is a passive, causal vacuum-response kernel with a normalized positive measure. Passivity floors the response amplitude at the horizon scale — it is the source of the scale-tie. But passivity has a mathematical consequence: $\sup K = 1$. A passive kernel can only **suppress**. On the RAR shell it delivers exactly ρ/ν ; deep-MOND $K \rightarrow 0$, the source dressed all the way down.

MOND-lensing demands the opposite. The phantom $(\nu - 1)\rho$ is an **enhancement** to $\nu\rho$, with $\nu > 1$ — strictly *outside* the passive cone whose supremum is 1. The needed uu -coefficient is $\nu - 1/\nu = 1/\sqrt{y(y+1)}$ (unbounded, deep

$\sim 1/\sqrt{y}$); the passive kernel supplies $1/\nu$; the shortfall $\nu^2 - 1 = 1/y$ **diverges** as $y \rightarrow 0$.

So the dichotomy is exact and structural, not a failed search:

The very property that ties a_0 's scale to the vacuum — passivity, $\sup K = 1$ — is the property that forbids the lensing phantom. Enhancement requires an anti-dissipative (pumped) kernel, whose amplitude is a free coupling, so a_0 would no longer be cH_Λ/Z .

The nonlocality lemma closes the loop from the other side: the mass-correct phantom of a point mass is an unbounded halo ($M_{\text{ph}}(r)/r \rightarrow \sqrt{a_0 M/G}$), so sourcing it universally forces a nonlocal carrier with a spatial-kinetic term — a propagating dof whose acceleration scale is a free coupling. Passivity forbids the enhancement; the halo's nonlocality forces propagation; propagation frees a_0 . The two horns are the same wall seen from two sides.

Two honest guards on this statement. First, boundedness ($\neg L$) uses **only** $\sup K = 1$ — it does not invoke the a_0 value-derivation at all. So “ a_0 derived” and “ $\neg L$ ” are not one claim chained back on itself; they are two consequences of the **one** passivity premise (a common-premise entailment, not a biconditional — we do not claim, and do not need, the converse “ a_0 -derived $\Rightarrow$ passive”). Second, this is *sufficiency*, not necessity: passivity $\Rightarrow$ (a_0 -scale-tied $\wedge \neg L$). Whether some **non-passive** kernel could also derive a_0 — breaking the lock — is precisely the open pump door of Section 7, so the lock itself is conditional on staying inside the passive class.

7. Scope and conditionality

We are explicit, both ways, about the strength of this result.

It is conditional. The obstruction rests on the single premise of MI-kernel passivity: the response is bounded, causal, positive-measure (Herglotz–Nevanlinna), $\|K\| \leq 1$, with the sum rule $\int d\mu/|t| = 1$, so its real-space dressing obeys $\sup K = 1$. This premise is banked and frozen (MI action v4, v11) but it is a *premise*, not a metatheorem. Were the kernel not passive, the argument would not run.

Within the passive class, D and $\neg L$ share one origin. The same passivity premise is what *derives* a_0 's scale and what caps K at 1; $\neg L$ follows from the cap alone (Section 6). One cannot, inside the passive class, keep the derivation while discarding the boundedness that forbids L — they are two faces of the one hypothesis, which is what makes the obstruction a real dichotomy rather than a coincidence of two separate assumptions. We state this as common-premise entailment, not as a biconditional lock: the claim is passivity $\Rightarrow$ (D -scale $\wedge \neg L$),

and whether D could ever hold *without* passivity is the open door below. So the lock is genuine within the passive class and itself conditional on it.

The one MI escape, and why it is currently shut. The only way out that keeps a_0 tied to cH_Λ with a *derived* amplitude is a causal **enhancing** kernel ($\nu > 1$) — an anti-dissipative / pumped vacuum response, i.e. a *non-passive* kernel that nonetheless derives a_0 . The banked sign walls (v4, v11) find no such channel pump-free: passive, reactive, and 4-point/dS-positivity edges are all sign-blind or anti-MOND-locked, and the enhancement reopens only under a *new forced pump mechanism* that has not been exhibited. Absent that, the no-go stands as effectively final for the passive class. **We do not claim the theory is closed.** The door labeled “forced pump” remains open in principle; what the banked results establish is that it is not open by any pump-free route so far identified — and, since a non-passive a_0 -deriving kernel is exactly what would break the D /boundedness lock, that same door is the one place the lock could fail.

We do not overclaim in either direction. The obstruction is not more final than “conditional-on-passivity”; nor is it softened — within the passive class it is exact and realized three independent ways. There is no manufactured no-go and no manufactured completion.

8. Consequences

(i) **A completion exists — as modified gravity, at the cost of the a_0 derivation.** C2 and C3b constructively demonstrate a single-metric, ghost-free (C3b) or ghost-cost (C2), GW170817-safe field theory that reproduces MOND dynamics *and* MOND lensing, with the correct deep slope. This is real, honest content. But it is modified *gravity* in the Deffayet–Esposito–Farèse–Woodard / AeST (Skordis & Zlosnik 2021) class, not modified inertia: it forfeits the vacuum-derived $a_0 = cH_\Lambda/Z$, demoting a_0 to a free Lagrangian coupling, plus (C2) a separate cosmology fit and a contested would-be ghost, or (C3b) one new propagating scalar and a forfeited cosmology. The cost is precisely the program’s single most distinctive claim.

(ii) **Lensing becomes a shared-MG observable.** Because the only completions that reproduce the lensing phantom are MG (C2, C3b degenerate there with AeST), weak lensing can no longer discriminate MI from MG — both predict the same $F \rightarrow 1$ MOND-lensing relation. The clean modified-gravity-*impossible* discriminant reverts to the non-adiabatic **relational velocity-dispersion spread**: MI predicts a 6–13% spread in member dispersions from the relational (frame-history-dependent) inertia, while MG predicts *exactly* zero. That observable — currently underpowered — is the one place MI $\neq$ MG survives, and it is where discrimination effort should now go. (Any MG completion also inherits the AeST Q_2 -quadrupole Solar-System

caveat of Desmond, Hees & Famaey 2024 and Park et al. 2026 — a shared-MG cost, honestly stated.)

(iii) **The obstruction shows the a_0 reframing is load-bearing.** If $a_0 = cH_\Lambda/Z$ were a trivial or cosmetic relabeling, one could keep it while adding any lensing structure. That it is *incompatible* — within the passive class — with single-metric MOND-lensing shows it is a genuine, restrictive, physical commitment: the passivity that derives its scale is a real constraint on the theory space. The reframing’s distinctive content is exactly what is forbidden on the MG-lensing side. Far from a funeral, this is evidence the reframing is doing work.

9. Conclusion

We have identified, and machine-verified on both footings, an obstruction — conditional on kernel passivity — in the de Sitter–Unruh modified-inertia program: **a passive-kernel-derived a_0 and single-metric MOND weak-lensing cannot coexist.** The central implication is $D \wedge S \Rightarrow \neg L$; equivalently $\{D, S, L\}$ is the unique minimal unsatisfiable subset of $\{D, S, G, L\}$, with $\{D, S\}$ = pure MI, $\{S, L\}$ = Deffayet–Esposito–Farèse–Woodard / AeST (a_0 free), and $\{D, L\}$ = a disformal two-cone construction (a_0 derived) killed observationally by GW170817. Ghost-freedom is not the lever. The mechanism is a passivity/amplification dichotomy — the passive kernel that floors a_0 ’s scale at the horizon can only suppress the source to ρ/ν , whereas lensing needs enhancement to $\nu\rho$, outside the passive cone — reinforced by a nonlocality lemma that turns any mass-correct lensing carrier into a propagating degree of freedom whose acceleration scale is a free coupling. Three explicit candidates witness the surviving corners: C1 fails lensing by mass-blindness ($\sim 2.4\%$ of the phantom on a cluster fixed to a galaxy, $\sim 41\times$ under-lensed), while C2 and C3b close lensing exactly but only as modified gravity with a_0 free.

The result is conditional on kernel passivity — but that premise is the very thing that derives a_0 ’s scale, so within the passive class the derivation and the obstruction are two faces of one hypothesis. The only MI escape, a causal enhancing kernel, is shut by the banked sign walls pump-free and reopens only under a new forced pump mechanism not yet exhibited; we therefore do not claim the theory is closed. The honest map: the MI program is complete for dynamics (0.108 dex on SPARC) and viable in cosmology, and its lensing sector completes only by crossing into modified gravity at the cost of the a_0 reframing. Lensing is now a shared observable; the clean discriminant is the relational σ -spread. And the obstruction itself is the strongest evidence that the $a_0 = cH_\Lambda/Z$ reframing is load-bearing physics rather than a relabeling.

Appendix A. Verification scripts

All scripts are committed in `prep_2026/mi_lensing_completion/`, run against a frozen read-only repository, and exit 0 on both footings ($a_0 = 9.36 \times 10^{-11}$ canonical = cH_Λ/Z , and 1.13×10^{-10} alternate). They use sympy for exact symbolic checks and numeric grids where a radical will not auto-collapse. No candidate passes all checks, so none is asserted as a full completion.

- **c1_frame_curvature.py** (11/11, exit 0). The frame-curvature variation gives $M_{\text{ph}}(r) = (c^2/G)\kappa r^2 F'(r)$; lensing closure forces $F'(y) \propto \sqrt{M}$ (sympy-exact); the two-mass ratio $\sqrt{M_2/M_1}$ is y -independent (mass-blindness); the cluster phantom fraction $\sqrt{6 \times 10^{10}/10^{14}} \approx 0.024$ ($\sim 41 \times$ under-lensing); and the honest-variation Ostrogradsky term ($\square F(K)$ contains Φ''').
- **c2_nonlocal.py** (18/0, exit 0). The Deffayet–Esposito–Farèse–Woodard nonlocal-in-matter term sources $\nu\rho$ with zero anisotropic stress, no-slip $\Phi = \Psi$, $F \rightarrow 1$ with the $\sqrt{a_0 g_{\text{bar}}}$ slope; $c_\gamma = c_{\text{GW}}$ (nonlocal prefactor cancels in the GW dispersion); the $\pm \frac{1}{2}$ would-be-ghost eigenvalues; and form-invariance of ν under $(a_0, g_{\text{bar}}) \rightarrow \lambda(\dots)$ (a_0 free — noted as *shared* with pure MI, so structural, not the primary marker).
- **c3_carrier.py** (exit 0). C3a delivers the $1/\nu$ suppression (shortfall $\nu^2 - 1 = 1/y$); C3b’s local $h \sim a_0/|a|$ (AQUAL) gives the unbounded halo $M_{\text{ph}}/r \rightarrow \sqrt{a_0 M/G}$, forcing a propagating scalar with a_0 free.
- **nogo.py** (exit 0). The completeness check: the nonlocality limit; the boundedness shortfall and the tension-signed slip leg $2K'X/K = 1/(2y + 1)$; the mass-blindness ratio; footing independence of the wedge; and the finite truth table over $\{D, S, G, L\}$ showing $D \wedge S \Rightarrow \neg L$, that $\{D, S, L\}$ is the unique minimal unsatisfiable subset (each proper subset realized, including the GW-dead $\{D, L\}$ disformal witness), and that the target corner and its ghost-relaxed sibling are both absent, with the surviving corners mapped onto C1/C2/C3. The table is bookkeeping over Lemmas 1–2, not an exhaustion over all Lagrangians.
- **verify_adversarial.py** (exit 0). An independent re-derivation (not reusing the candidate assertions) of the wedge ν^2 , the passivity/amplification dichotomy, the mass-blindness $\sqrt{M}$, the frame-lock fork, a direct passive-kernel counterexample search, and a check that C2’s $F \rightarrow 1$ is a genuine (not manufactured) closure that is nonetheless MG with a_0 free. Trap sweep both directions: no manufactured completion, no manufactured no-go, no hidden ghost in a “winner,” $c_\gamma = c_{\text{GW}}$ preserved, Cassini/cosmology honest. Verdict: **UPHELD**.

References

- Abbott, B. P., et al. (LIGO/Virgo, Fermi-GBM, INTEGRAL) (2017). Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A. *ApJ Lett.* **848**, L13. arXiv:1710.05834. [The $|c_\gamma/c_{\text{GW}} - 1| \lesssim 10^{-15}$ bound excluding the disformal second cone.]
- Bekenstein, J. D. (2004). Relativistic gravitation theory for the modified Newtonian dynamics paradigm. *Phys. Rev. D* **70**, 083509. [TeVS; the two-metric / disformal lensing lineage.]
- Brouwer, M. M., et al. (2021). The weak lensing radial acceleration relation: constraining modified gravity and cold dark matter theories with KiDS-1000. *A&A* **650**, A113. arXiv:2106.11677; DOI 10.1051/0004-6361/202040108. [KiDS-1000 isolated-galaxy lensing RAR; the $\sim 27\sigma$ benchmark.]
- Deffayet, C., Esposito-Farèse, G., & Woodard, R. P. (2011). Nonlocal metric formulations of MOND with sufficient lensing. *Phys. Rev. D* **84**, 124054. arXiv:1106.4984. [The C2 nonlocal-in-matter, purely-metric MOND-lensing realization.]
- Desmond, H., Hees, A., & Famaey, B. (2024). On the tension between the radial acceleration relation and Solar System quadrupole in modified gravity MOND. *MNRAS* **530**, 1781.
- Milgrom, M. (1983). A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *ApJ* **270**, 365.
- Milgrom, M. (1999). The modified dynamics as a vacuum effect. *Phys. Lett. A* **253**, 273. astro-ph/9805346. [The identical ν -kernel modified-inertia law, Eq. 9; the framework’s distinctive content is the cH_Λ/Z coefficient, his was $2cH_\Lambda$.]
- Park, M., et al. (2026). Solar-System quadrupole constraints on modified gravity. arXiv:2602.17884.
- Skordis, C., & Zlosnik, T. (2021). New relativistic theory for modified Newtonian dynamics. *Phys. Rev. Lett.* **127**, 161302. arXiv:2007.00082. [AeST; the single-metric MG comparison class.]
- *Background (not the C2 citation)*: Deffayet, C., & Woodard, R. P. (2009). Reconstructing the distortion function for nonlocal cosmology. *JCAP* **08**, 023. arXiv:0904.0961.
- **Self-citations.** Zimmerman, C. (2026). MI Field Theory Results 2026, DOI 10.5281/zenodo.21403470 (erratum 21415677); The MI Action v4, DOI 10.5281/zenodo.21264727, v11 21284144; the covariant / cluster no-go, DOI 10.5281/zenodo.20779562.

Reproduce: `python3 {c1_frame_curvature,c2_nonlocal,c3_carrier,nogo,verify_adversarial}.py` in `prep_2026/mi_lensing_completion/` — all exit 0. Both a_0 footings carried throughout; the verdict is footing-independent. This is a conditional obstruction result, not a closure of the program.